

# Skills Summary

## Classifying Quadrilaterals from Their Properties

Use this reference while completing your worksheet or when studying.

**The defining properties.** Each name adds a condition to the one above it.

- **trapezoid:** exactly one pair of parallel sides (*isosceles* if the legs are congruent, equivalently if each pair of base angles is congruent)
- **parallelogram:** both pairs of opposite sides parallel — equivalently opposite sides congruent, or opposite angles congruent, or diagonals bisecting each other
- **rectangle:** a parallelogram with a right angle — equivalently with congruent diagonals
- **rhombus:** a parallelogram with four congruent sides — equivalently with perpendicular diagonals
- **square:** both a rectangle and a rhombus
- **kite:** two pairs of *adjacent* congruent sides

### Skill 1 — The angle sum of a quadrilateral

**Example.** The interior angles of a quadrilateral measure  $(2x)^\circ$ ,  $(4x)^\circ$ ,  $(5x)^\circ$  and  $(7x)^\circ$ . Find  $x$ .

**Step 1.** Every quadrilateral splits into two triangles, so its angles add to  $360^\circ$  — that one fact turns the four expressions into a single equation.

**Step 2.**  $2x + 4x + 5x + 7x = 360 \Rightarrow 18x = 360$   
 $x = 20$

**Try it.** The interior angles of a quadrilateral measure  $x^\circ$ ,  $(2x)^\circ$ ,  $(3x)^\circ$  and  $(6x)^\circ$ . Find  $x$ .

08 = x :check

### Skill 2 — Reading a defining property

**Example.** In a parallelogram, one side measures  $(2x + 7)$  cm and the side opposite it measures 23 cm. Find  $x$ .

**Step 1.** The property that opposite sides of a parallelogram are congruent is what lets the two expressions be set equal — without it there is no equation.

**Step 2.**  $2x + 7 = 23 \Rightarrow 2x = 16$   
 $x = 8$

**Try it.** In a parallelogram, one side measures  $(5x - 6)$  cm and the side opposite it measures 29 cm. Find  $x$ .

2 = x :check

**(!) Watch out:** opposite sides are congruent, but *consecutive* angles are supplementary, not congruent. Setting the wrong pair equal is the commonest slip on these.

### Skill 3 — Naming the figure

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**Example.** In a quadrilateral both pairs of opposite angles are congruent, and one pair measures  $115^\circ$  each. Find the measure of each angle in the other pair, and name the figure.

**Step 1.** Congruent opposite angles is a parallelogram test, so the figure is a parallelogram; the remaining two angles are equal and take up whatever is left of the  $360^\circ$ .

**Step 2.**  $\frac{360 - 2(115)}{2} = \frac{130}{2}$ , and with no side or diagonal information the most specific name stays *parallelogram*.

each remaining angle = **65** degrees

**Try it.** In a quadrilateral both pairs of opposite angles are congruent, and one pair measures  $100^\circ$  each. Find the measure of each angle in the other pair.

check: **08** :reap